CNN Architectures: A Simplified Guide

1. AlexNet (2012) — "CNNs actually work"

Why it mattered

First CNN to dominate image classification (ImageNet). Proved deep learning was practical.

- **Key idea:** Use CNNs + ReLU + dropout + GPUs.
- **Think of it as:** The model that opened the door.

2. VGGNet (2014) — "Deeper is better"

What changed

Much deeper than AlexNet. Uses many 3×3 convolutions instead of large filters.

- **Why that's good:** Same receptive field, more non-linearities, fewer parameters, and a cleaner design.
- **Think of it as:** Stacking small Lego blocks instead of big ones.

3. Inception / GoogLeNet (2014) — "Deep AND efficient"

Problem

VGG was accurate but huge and computationally expensive.

- **Key idea (Inception module):** Run $1 \times 1, 3 \times 3, 5 \times 5$ convolutions in parallel. Use 1×1 convolutions as bottlenecks to reduce cost.
- **Result:** Very deep (22 layers) with very few parameters.
- **Think of it as:** Looking at the image in multiple ways at the same time, but cheaply.

4. ResNet (2015) — "Fixing deep training"

Problem

Making networks deeper sometimes made accuracy worse due to vanishing gradients, not overfitting.

- **Key idea (Skip connections):** Learn $F(x) + x$ instead of $F(x)$.
- **Why it works:** Gradients flow easily; can train 100+ layers.
- **Think of it as:** Shortcuts so the network doesn't get lost.

5. EfficientNet (2019) — "Scale smart"

Compound Scaling

Scale depth, width, and image resolution together using a single coefficient (ϕ).

- **Pros:** Higher accuracy with far fewer parameters and FLOPs.
- **Think of it as:** Growing the network evenly in all directions.

6. MobileNet — "Run CNNs on phones"

- **Key idea:** Depthwise-separable convolutions (Depthwise $\rightarrow$ per channel; Pointwise $\rightarrow$ mix channels).
- **MobileNet v2:** Adds residual connections and expansion layers.
- **Think of it as:** Doing the same job for much cheaper computation.

Comparison Table

Model	Main Goal	Key Innovation
AlexNet	Make CNNs practical	ReLU, dropout, GPU training
VGG	Improve accuracy	Very deep, small 3×3 filters
Inception	Reduce computation	Parallel multi-scale filters
ResNet	Train very deep nets	Skip (residual) connections
EfficientNet	Optimal scaling	Compound depth/width/resolution
MobileNet	Mobile efficiency	Depthwise-separable convs

Memory Hooks

$\rightarrow$ **AlexNet:** CNNs work
$\rightarrow$ **VGG:** Deeper with small filters
$\rightarrow$ **Inception:** Parallel multi-scale
$\rightarrow$ **ResNet:** Skip connections
$\rightarrow$ **EfficientNet:** Smart scaling
$\rightarrow$ **MobileNet:** Lightweight CNNs